

The Utility Problem

Many utilities today publish details of their electricity power systems on public websites for reasons of DER (Distributed Energy Resource) expansion, transparency, policy compliance, customer engagement, etc.

However, the quality, granularity, and usability of these public maps vary significantly across utilities. Some provide static PDFs updated quarterly, while others offer live web maps with filterable layers and customer interaction. The lack of standardization, challenges in maintaining data accuracy, and the labor-intensive process of producing these maps manually represent major barriers to scaling these efforts. Furthermore, the underlying GIS and operational datasets must be validated and synchronized to avoid the risk of publishing outdated or misleading information.

As grid modernization and DER proliferation continue, the demand for accurate, up-to-date public visibility into the power grid will only increase. This presents a growing challenge—and opportunity—for utilities to automate and standardize grid data publication through a repeatable, scalable digital toolset.

The Data Solutions and Methods

The Awesense Platform offers utilities a streamlined path to address this challenge. With its robust data ingestion, validation, and visualization capabilities, the platform can act as a central engine for publishing and maintaining high-quality power grid maps for public access.

Once GIS, connectivity, and relevant operational data (such as feeder or transformer load, outage zones, or renewable hosting capacity) are ingested into the Awesense Energy Data Model (EDM), utilities can use the platform's tools to create filtered, permission-controlled map views. These views can be tailored to show specific asset classes, voltage levels, or segments (parts) of the grid. Sensitive internal data can be automatically excluded or anonymized where needed.

Key features of this use case include:

- **Automated Publishing Pipelines:** Using Awesense APIs, map tiles or full embedded maps can be generated for websites and updated automatically daily, weekly, or event-driven.
- **Custom Layer Control:** Utilities can selectively expose various layers that can be visualized in different formats.
 - One of these formats can be heatmaps, which provide intuitive, color-coded insights into the state of the grid.
 - Some analyses can be created directly in TGI using Segment Analyzer.
 - Examples:
 - **Load heatmaps** show real-time or recent feeder and transformer loading levels.
 - Some analyses can be created using APIs and tools such as QGIS or other mapping tools, as detailed in our [blog post](#). And then they can be visualized directly in TGI.

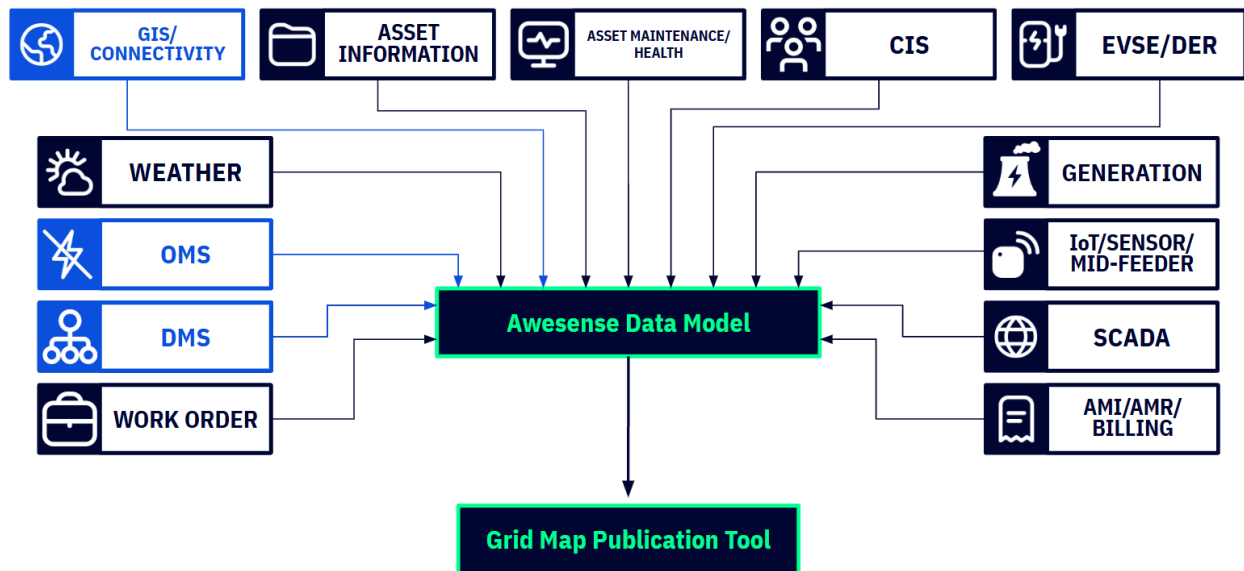
- Examples:
 - **Capacity heatmaps** display the available headroom or remaining hosting capacity on feeders or substations.
 - **DER heatmaps** visualize the number or density of connected distributed energy resources across different areas.
- Another format could be an outage map. Outage visualization layers can display current outages in the service area, including affected zones and counts of impacted customers. This information is used to display status and public awareness and is obtained from the outage management system.

All layers can be interactively toggled and filtered by end users, ensuring a clear and engaging public-facing interface while maintaining internal control over visibility and content.

This process transforms what was once a slow, manual map-making effort into a dynamic, validated, and maintainable interface between the utility and its stakeholders—regulators, customers, and others.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. Please contact [Awesense](#) if you are interested in the implementation of this use case.